Corporate Finance — Problem Set 2

Answer key (full derivations in `PS2-solutions.pdf`)

Exercise 1 (slump, boom; probabilities $\frac{1}{2}$)

(a) Stock: $2.5\psi_b = 1.05 \Rightarrow \psi_b = 0.42$. Bond: $\psi_s + \psi_b = 0.92 \Rightarrow \psi_s = 0.50$. ($r_f = 8.70\%$; project value $A = 0.5(0.50) + 3(0.42) = 1.51$, NPV $= 0.51$.)

(b) $C(1) = 2(0.42) = 0.84$; $C(1.5) = 1.5(0.42) = 0.63$.

(c) $P(1) = 0.5(0.50) = 0.25$; $P(1.5) = 1(0.50) = 0.50$.

(d) $0.84 + 0.92 - 0.25 = 1.51 = 0.63 + 1.38 - 0.50$. Put–call parity: $C(K) + K/(1 + r_f) - P(K) = A$ for any K. With $K =$ face value, equity is the call and risky debt is a safe bond minus a put, so $E + D = A$ whatever the face value: MM I.

(e) Debt of 1 is risky ($1/0.92 > 0.5$): $0.25 + 0.42F = 1 \Rightarrow F = 1.7857$. $A = 1.51$, $D = 1$, $E = 0.51$; $r_A = 1.75/1.51 - 1 = 15.89\%$, $r_D = (0.25 + 0.5F)/1 - 1 = 14.29\%$, $r_E = 0.5(3 - F)/0.51 - 1 = 19.05\%$.

(e′) $F = D/0.92$ for $D \leq 0.46$ (safe debt); $F = (D - 0.25)/0.42$ for $D > 0.46$. $A = 1.51$, $r_A = 15.89\%$ and $E = 1.51 - D$ for every D, and the entrepreneur always keeps 0.51. Safe region: $r_D = 8.70\%$, $r_E = r_A + \frac{D}{E}(r_A - r_D)$ rises from 15.89% to 19.05%. Risky region: $r_D = (0.25 + 0.5F)/D - 1$ rises to 14.29% at $D = 1$, and $r_E = 0.5/0.42 - 1 = 19.05\%$ stays flat, because equity then pays only in the boom. Full table: Appendix A of the detailed solution.

(f) $F = (1.1 - 0.25)/0.42 = 2.0238$. Firm value after the dividend: $A = 1.51$ ($D = 1.1$, $E = 0.41$). The entrepreneur has $0.1 + 0.41 = 0.51$, the same as before: **not better off**. The dividend only moves part of her wealth to $t = 0$.

Exercise 2 (millions of dollars)

(a) $r_U = 3\% + 1.2(5\%) = 9\%$ before. After (continuous rebalancing, $r_{TS} = r_U$): $r_E = 9\% + 0.25(9\% - 3\%) = 10.5\%$.

(b) $r_{WACC} = 0.8(10.5\%) + 0.2(3\%)(0.75) = 8.85\%$.

(c) $V_U = 25/0.09 = 277.78$, price 27.78. $V_L = 25/0.0885 = 282.49$, $V_{TS} = 4.71$. Price after the announcement 28.25. Debt $= 0.2(282.49) = 56.50$ buys back 2.00 million shares; 8.00 million remain at 28.25.

(d) Interest $= 0.2(25) = 5$, shield $= 1.25$ per year with FCF risk: $V_{TS} = 1.25/0.09 = 13.89$.

(e) $V_{TS} = \tau_c D = 25$; $V_L = 302.78$; price 30.28; $100/30.28 = 3.30$ million shares repurchased; 6.70 million remain.

Exercise 3

(a) $V_U = 80/1.15 = 69.57$ (NPV 9.57).

(b) Creditors need $60(1.066) = 63.96$; debt is risky: $0.5(60) + 0.5F = 63.96 \Rightarrow F = 67.92$. $E = 69.57 - 60 = 9.57$.

(c) In default creditors get $60 - 0.07(60) = 55.8$: $0.5(55.8) + 0.5F = 63.96 \Rightarrow F = 72.12$. $V_L = (80 - 2.1)/1.15 = 67.74$, $E = 7.74$. **Shareholders pay**: equity falls by $1.83 = PV(BC)$. Creditors price the loss in through the higher face value.

Exercise 4

(a) $D/E = 0.4/0.6$, $\beta_D = 0$: $\beta_E = 1.2(1 + 2/3) = 2.0$.

(b) $r_E = 12.5\% + \frac{2}{3}(12.5\% - 5\%) = 17.5\%$.

(c) Price $= 1.50 \times 14 = £21$. Interest per original share $= 0.05(0.4 \times 21) = 0.42$; $EPS = (1.50 - 0.42)/0.6 = £1.80$. **No benefit**: the price stays at £21; the higher EPS only compensates for the higher risk.

(d) $P/E = 21/1.80 = 11.67$ (from 14). Reasonable: riskier earnings are worth less per pound.